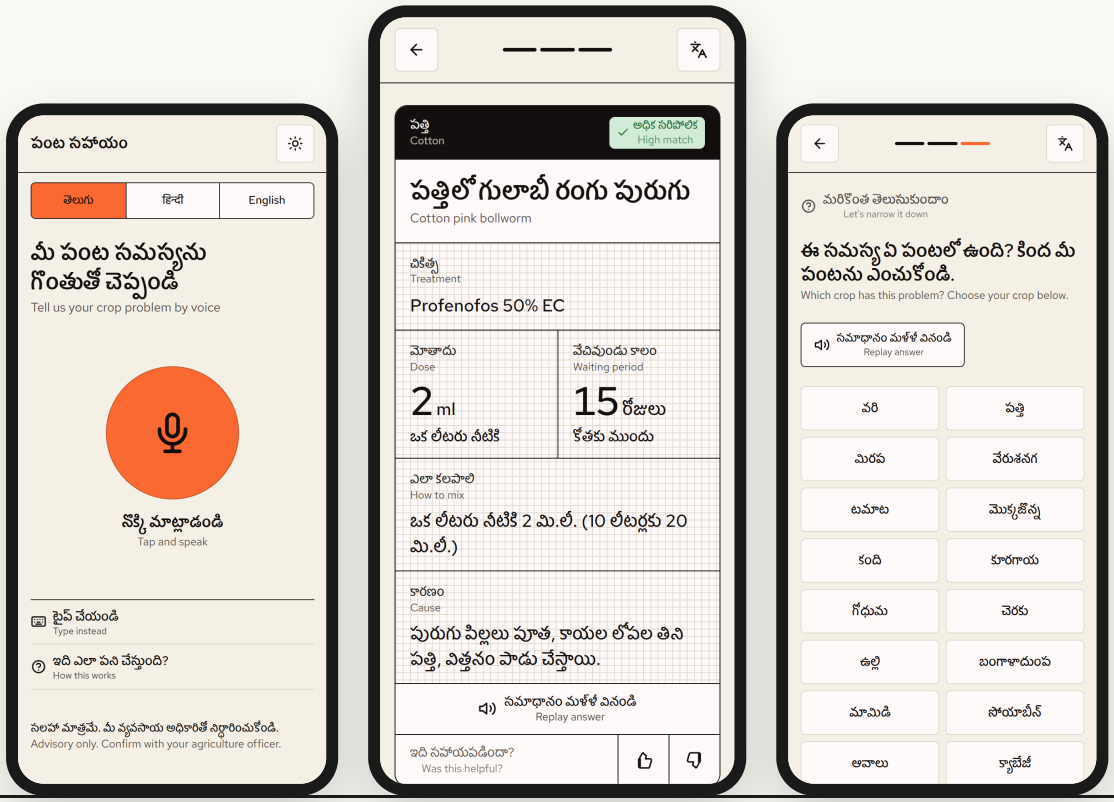
GNANI AI · GREAT INDIAN AI INTERNSHIP CHALLENGE

Pantha Sahayam

పంట సహాయం · फसल सहायता · Crop Help

A farmer **speaks** the crop problem in Telugu, Hindi or English, and **hears** back the right registered spray, the right dose and the safe waiting period, powered by **Gnani Prisma** and **Gnani Timbre**.



3
languages, voice in & voice out

41
checked problems on 16 crops

2,243
official CIB&RC registered uses

0
doses written by AI (every one is official)

PROJECT SUBMISSION

01 The problem

When a pest or disease appears in the field, a small farmer must decide **what to spray, how much, and when it is safe to harvest**, often within days and often alone.

> **Most small farmers decide what to spray by asking the pesticide shop, the same place that sells the spray.**

What goes wrong today

- **Wrong product.** A spray that isn't registered for that crop or pest wastes money and doesn't work.
- **Wrong dose.** Overdosing harms the farmer's health and the soil. Underdosing breeds resistant pests.
- **Spraying too close to harvest.** Chemical residues reach the food on our plates.
- **Help is far away.** Agriculture officers are few, and calling a helpline means explaining in a language and vocabulary that may not be the farmer's own.

Why existing apps don't reach them

- **Text-heavy and English-first,** while many farmers are more comfortable speaking than typing, in Telugu or Hindi.
- **Sign-ups, OTPs and installs,** which are real barriers on a basic phone with limited data.
- **Generic AI chatbots** can sound confident and still *invent* a pesticide dose. On a farm, a made-up number is dangerous.

Who we are building for

Small and marginal farmers, who hold the large majority of India's farm holdings (about 86% according to the Agriculture Census 2015-16). We start with farmers in **Andhra Pradesh and Telangana (Telugu)** and the **Hindi-speaking states**, with English for extension workers and students. The phone is already in their pocket; what's missing is help that **listens and speaks their language** and can be **trusted**.

Problem statement

How might a small farmer, using only their voice and their own language, get pesticide advice that is correct, officially registered and safe, in under a minute, without an app install or an account?

02 Our solution

Pantha Sahayam ("crop help") is a voice-first web app. The farmer opens a link, taps one big microphone button and describes the problem in their own words. Gnani's speech models understand them and speak the answer back.

VOICE FIRST

Speak, don't type

One big mic button. Up to 30 s of natural speech in Telugu, Hindi or English. Typing is optional.

TRUSTED

Official doses only

Every product, dose and waiting period comes from the Government of India's CIB&RC registration list, and is checked automatically.

NO BARRIERS

No login, no install

It works in any phone browser through a link. No account, no OTP, no app store.

What the farmer gets

Feature	What it means for the farmer
Answer in text + voice	The problem's name and cause, the product, the **dose per litre**, how to mix 10 litres and the **days to wait before harvest**, spoken aloud by Gnani Timbre.
Change language anytime	A small language key at the top re-reads the same answer, text and voice, in Telugu, Hindi or English. Useful when a son or daughter, an officer or a shopkeeper is helping.
"You said..." check	Before answering, the app shows what it heard. The farmer can fix it, so a mis-heard word never becomes a wrong spray.
No dead ends	A broad question like *"there are caterpillars on my plant"* leads to **"which crop?"**, then that crop's caterpillar problems as buttons, then the answer.
Waiting period by crop	The same spray can need 3 days on tomato but 29 days on pigeon pea. The app uses the farmer's crop, and the longest period when the crop is unknown.
Other registered sprays	Alternatives registered for that crop and pest, each linked to its page in the official government PDF.
Safety net	"Draft advice: confirm with an officer" until an expert verifies each entry, plus a one-tap call to the free **Kisan Call Centre 1800-180-1551**.

Our key design decision: AI listens and speaks, but never invents advice

We use Gnani's models for what they do best, **understanding and producing Indian-language speech**. The *advice itself* comes from a checked knowledge base tied row by row to the government registry. A wrong pesticide dose can hurt a family, so we chose safety over a free-form chatbot.

03 How it works

1
Speak
The farmer taps the mic and describes the problem. The browser records and converts it to 16 kHz WAV.

2
Gnani Prisma
Speech → text in Telugu, Hindi or English.

3
Confirm
"You said…": the farmer confirms, edits or re-records.

4
Understand
A symptom matcher scores 41 checked problems from the symptom words, crop and kind of problem.

5
Gnani Timbre
The answer is spoken back, and the text stays on screen.

✅ **One clear problem**
Full answer: product, dose, waiting days, other registered sprays.

❓ **Partly sure**
One yes/no question, e.g. "Are there eye-shaped spots with grey centres?"

🌾 **Broad question**
"Which crop?" → that crop's problems as buttons → answer. If none fits, the officer.

Inside the "understanding" step

- **Symptom words in three languages**, as farmers actually say them (e.g. గులాబీ రంగు పురుగు, गुलाबी सुंडी, "pink bollworm").
- **Crop awareness:** evidence for the right crop raises the score, and a crop mismatch lowers it.
- **Tolerant of speech slips:** fuzzy matching handles what real speech-to-text returns, like "bowl worm" for bollworm, which we saw from Prisma during testing.
- **Explainable:** transparent rules and weights rather than a black box, so an officer can see why a problem was chosen.

Architecture (one deployable service)

Layer	What runs there
Phone browser	React app: screens, mic recording and live level meter, 16 kHz WAV conversion, audio playback, language switch
API server	Flask + Waitress: `/api/advise`, `/api/answer` (language switch), `/api/feedback`, `/api/health`. The same server serves the React app.
Intelligence	Symptom matcher · guided crop and problem choices · answer templates in 3 languages · dose guard · registry lookups
Data	`knowledge_base.json` (41 problems × 3 languages) · `pesticides.db` (2,243 CIB&RC uses, SQLite)
Gnani AI	Prisma (speech → text) and Timbre (text → voice), called only from the server, so the API key never reaches the phone
Hosting	One Docker image (React build + Python server). Deploys on Vercel or Render as a single https link.

04 Gnani models used

Model	Role in Pantha Sahayam	Integration
Gnani Prisma v2.5 **speech-to-text**	Understands the farmer's spoken problem in **Telugu (te-IN), Hindi (hi-IN) and English (en-IN)**, the core input of the app.	`POST api.vachana.ai/stt/v3`: multipart WAV + language code; clips up to 30 s. `prisma-v3.0` is one setting away.
Gnani Timbre v3.0 **text-to-speech, voice "Lakshmi"**	Speaks **every answer, follow-up question and choice prompt** in the farmer's language, including when the language is switched.	`POST /api/v1/tts/inference`, returning WAV (22,050 Hz). Timbre v2.5 voices (Lavanya, Nalini, Kaveri) can be chosen by setting.

Tested live, not just integrated

- Round trips Timbre → Prisma reproduced Telugu, Hindi and English sentences word for word.
- A full voice question in Telugu (పత్తి కాయల్లో గులాబీ రంగు పురుగు ఉంది) was transcribed, matched and answered with Telugu voice. Switching the answer to Hindi and English produced new Gnani voice with identical doses.
- Real Prisma mishearings in English ("pink ballworm", "bowlworm") were captured and the matcher was made tolerant to them.
- The Gnani client handles all three documented error formats, retries once on temporary errors or rate limits, and is covered by tests that use recorded API shapes (no credits spent).

What about Evon (gnani-evon-v3.3-30B-A3B)?

We evaluated Gnani's Evon language model. Our position is a deliberate design choice:

- **Advice stays deterministic.** A language model can phrase advice beautifully and still change a number. For pesticide doses that risk is unacceptable, so Evon is *not* in the critical path.
- **Ready as an optional layer.** The code already supports an Evon deployment (OpenAI-compatible endpoint, `GNANI_LLM_URL`) to make answers more conversational. A **number guard** rejects any Evon output that adds, drops or changes a dose or waiting period, and falls back to the checked template.
- **Hosting reality.** Evon is a 30B-parameter model without a hosted API; it needs data-centre GPUs (A100/H100). We enable it when a hosted endpoint is available.

05 Technology stack

Frontend

Tech	Use
React 18	Screens as one state machine, back-button support
Vite 5	Build (~57 kB JavaScript after gzip)
MediaRecorder + Web Audio	Recording, live mic meter, silence detection, 16 kHz WAV
Web Speech API	Fallback recognition and voice
CSS design tokens	No UI framework; 44 px+ touch targets, sunlight mode, phone / tablet / desktop layouts
Hind Guntur · Hind · Red Hat	Proper Telugu and Devanagari rendering

Backend

Tech	Use
Python 3.12	API, matcher, data tools
Flask 3 + Waitress	API server, also serves the React build
SQLite	Official registry: 2,243 CIB&RC uses, read-only
JSON knowledge base	41 problems × 3 languages
requests · dotenv · CORS	Gnani calls, settings, origin allow-list
PyMuPDF (tool)	Reads the official PDFs table by table into SQLite

Quality & safety

Tool	Use
pytest	46 automated tests
audit_kb.py	Every dose vs. official rows; fails the build on drift
Rate limit	Per device, protects Gnani credits
Pilot log	Reference code on every answer, plus ratings

Deployment

Tool	Use
Docker	Multi-stage: Node builds UI → Python serves all
Vercel / Render	One https link; microphone works on phones
GitHub	github.com/ghosttech07/Farmer-app
Secrets	Gnani key only as a host environment variable

06 Where the answers come from

Trust is the product. Every number a farmer hears can be traced to a government document.

Source	What we use it for
CIB&RC, *Major Uses of Pesticides*	Central Insecticide Board & Registration Committee, Directorate of Plant Protection, Quarantine & Storage, Govt. of India (ppqs.gov.in), **as on 30.09.2026**. Every product, dose and waiting period, plus the full registry (insecticides, fungicides, bio-insecticides, bio-fungicides).
TNAU Agritech Portal	Tamil Nadu Agricultural University: symptom descriptions of pests and diseases.
Kisan Call Centre	1800-180-1551, the free national helpline used as the safety net.

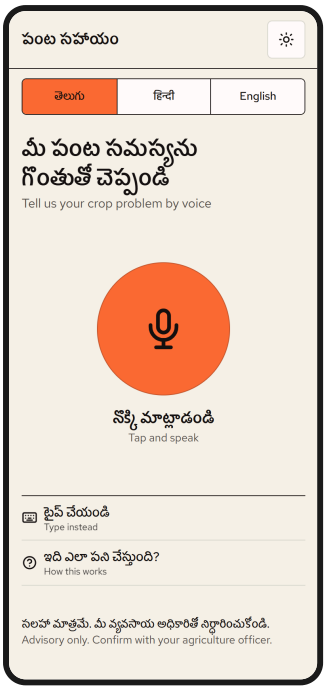
Five layers of protection

1. **One official row per treatment,** cited with the PDF page in the data.
2. **Automatic audit.** The product must be registered for that crop and pest, the dose must sit inside the official range, and the waiting period must not be shorter. During development this audit **caught two real errors**: a mancozeb formulation not registered for tomato early blight, and a neem product not registered against aphids. Both were corrected.
3. **Fixed answer templates** in each language, filled only from the checked entry, with a number guard on any rephrasing.
4. **Confirm before answer**, so a speech-to-text slip can be corrected by the farmer.
5. **Human sign-off.** Entries are marked *draft* until an agriculture officer verifies them. A pilot switch (`BLOCK_UNVERIFIED`) can send unverified entries to the officer instead.

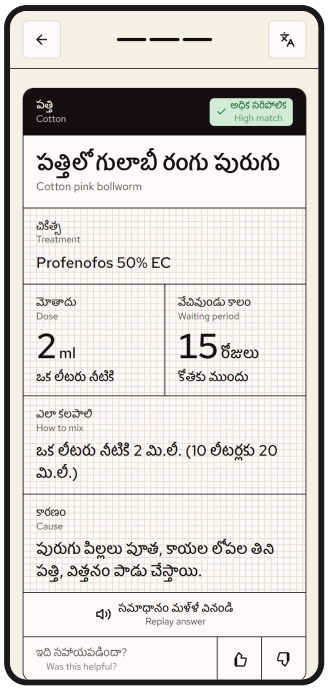
Coverage today

41 problems across 16 crops: rice, cotton, chilli, groundnut, tomato, maize, pulses, vegetables (brinjal, okra), wheat, sugarcane, onion, potato, mango, soybean, mustard and cabbage. They include pests (borers, armyworm, bollworms, leaf folder, aphids, whitefly, thrips, jassids, mites, hoppers, mealybug) and diseases (blast, sheath blight, blights, rusts, mildew, leaf spots, fruit rot). The knowledge base grows by adding entries; the audit checks each one against the registry automatically.

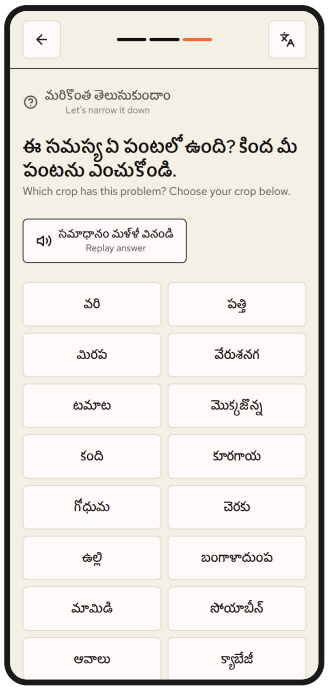
07 The app



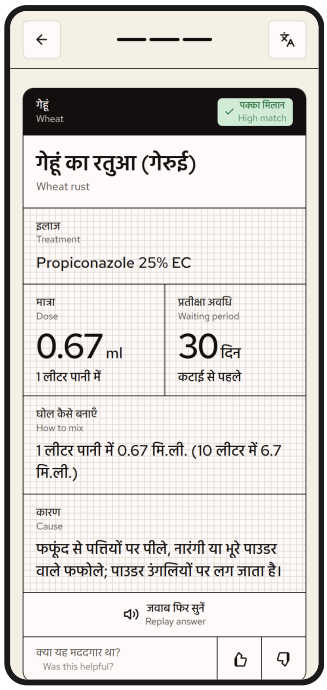
Home: one big mic, Telugu first, English underneath



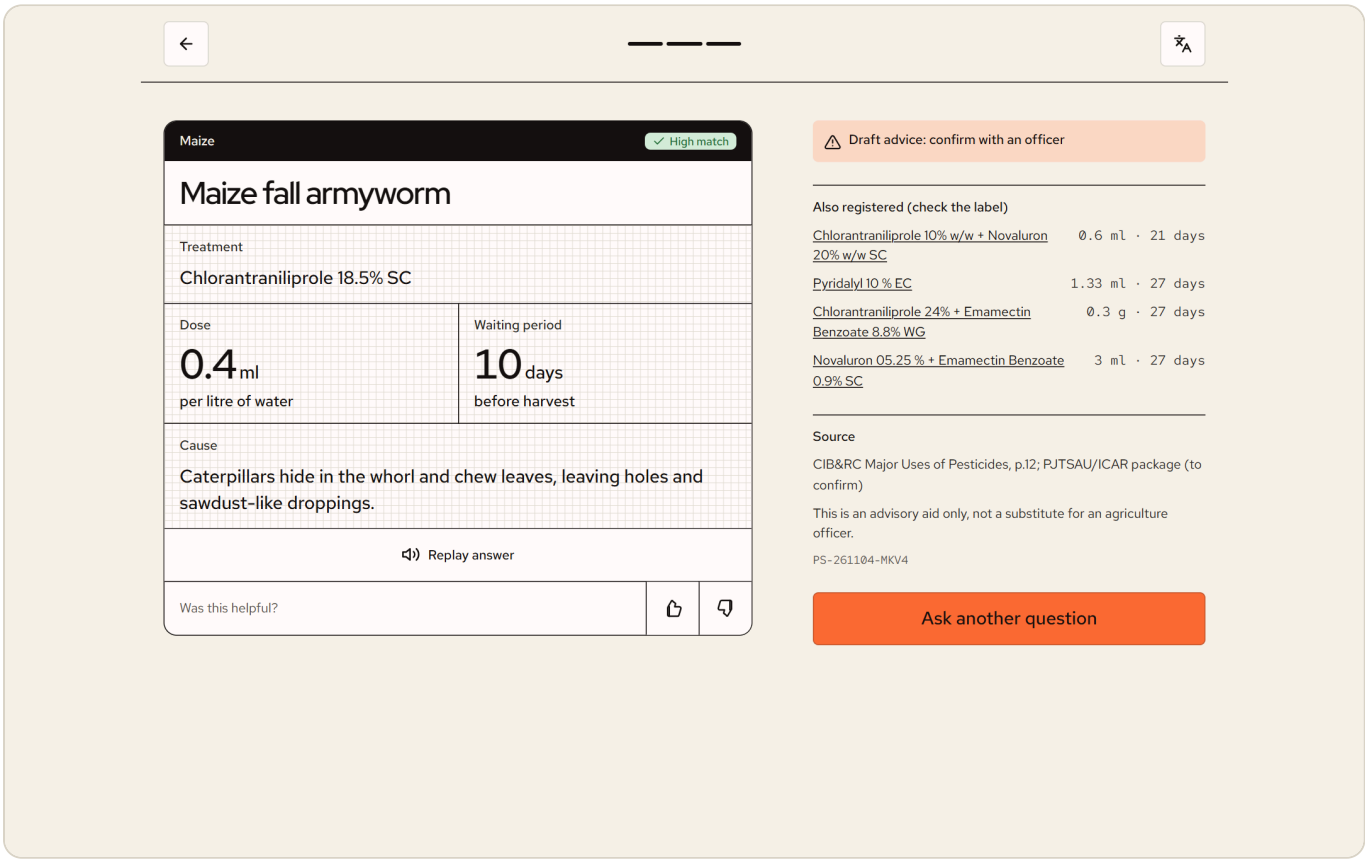
Answer: product, dose per litre, waiting days



Broad question → "which crop?"



Same flow in Hindi (wheat rust)



Desktop: maize fall armyworm, with other sprays registered by CIB&RC, each linked to its official page

08 Impact, novelty & roadmap

Impact

- **Right spray, right dose:** less money wasted on wrong or excess chemicals.
- **Safer food:** waiting periods spoken clearly, by crop.
- **Health:** less overexposure for farmers and families.
- **Inclusion:** works for farmers who prefer speaking over typing.
- **Officers' time:** common questions answered; the hard cases reach them.

What makes it different

- Voice in *and* voice out in Indian languages via Gnani.
- Every dose traceable to a government row, and audited.
- Language switch on any screen, mid-conversation.
- Guided choices instead of "I don't know".
- No login, no install: just a link.

Honest limits

- Answers cover the 41 checked problems; outside them the app guides the farmer to the Kisan Call Centre rather than guess.
- Entries await an agriculture officer's verification, and the Telugu and Hindi wording awaits native-speaker review.
- Field testing with real farmers in noisy conditions is the next step to tune the matcher.

Roadmap

Next	Plan
Pilot	10–15 farmers; the built-in pilot log and report measure answered rate, usefulness and what farmers did next
Coverage	Add the problems farmers actually ask about (from the pilot log); more Gnani-supported languages (Tamil, Kannada, Marathi…)
Expert loop	Officer review of entries; verified badge on answers
More help	Mandi crop prices (Agmarknet), weather-based spray timing, photo of the leaf as extra evidence
Evon	Conversational answers through a hosted Evon endpoint, kept behind the number guard

09 Demo video script (60 seconds)

Record on a phone using the deployed link (screen recording with sound). Speak the voice-over in English; the farmer lines are in Telugu and Hindi, and the app's Gnani voice plays as it is.

Time	On screen	Voice-over / what is said
0:00–0:06	Close-up of the home screen in Telugu, mic glowing	*"When pests attack, a small farmer has to decide what to spray, how much, and when it's safe to harvest. Usually alone."*
0:06–0:16	Tap mic; speak: పత్తి కాయల్లో గులాబీ రంగు పురుగు ఉంది	*"Pantha Sahayam lets them just speak, in Telugu, Hindi or English. Gnani Prisma turns the voice into text."*
0:16–0:22	"You said…" screen → tap *Yes, that's right*	*"The farmer checks what we heard, so a wrong word never becomes a wrong spray."*
0:22–0:34	Answer card: Profenofos 50% EC · 2 ml per litre · 15 days. Gnani Timbre speaks it in Telugu (let 3–4 s play)	*"The answer comes back as text and voice: the product, the dose per litre and the days to wait before harvest, all from the Government of India's CIB&RC registry."*
0:34–0:42	Tap the language key → हिन्दी; the Hindi voice plays	*"Switch language anytime, and Gnani Timbre speaks it again."*
0:42–0:53	New question: నా మొక్కకు గింగళి పురుగులు ఉన్నాయి → crop buttons → tap టమాట → tap the problem → answer	*"Even a vague question gets help: which crop, which problem, then the checked answer."*
0:53–1:00	Title card: logo, "Gnani Prisma + Timbre · 41 checked problems · every dose official"	*"Pantha Sahayam. Speak your problem. Hear the right answer. Built with Gnani AI."*

Recording tips

- Open the deployed link once before recording, since a free host may take a few seconds to wake up.
- Turn the phone volume up so Gnani's voice is audible in the screen recording.
- Record in a quiet room, holding the phone 20–30 cm from your mouth.

10 Presentation script (about 2 minutes)

Opening (problem). "Imagine a cotton farmer in Telangana who finds pink worms inside her cotton bolls. She needs to know what to spray, how much and when it's safe to pick, and today she asks the shop that sells the spray. Wrong products, double doses and spraying too close to harvest cost farmers money and health, and put residues on our food."

Why now. "She has a phone, but most farm apps are typed, English-first and need a login. She'd rather *talk*, in Telugu."

Solution. "Pantha Sahayam is a voice-first web app. She opens a link, taps one button and speaks. Gnani Prisma understands her Telugu, Hindi or English. She confirms what we heard. Then she gets the product, the dose per litre and the days to wait before harvest, on screen and spoken by Gnani Timbre. She can switch the language anytime. A vague question like 'caterpillars on my plant' becomes: which crop, which problem, answer."

Trust. "We made one hard choice: AI listens and speaks, but it never invents advice. Every dose comes from the Government of India's pesticide registration list, 2,243 official uses that we turned into a database, and an automatic audit checks every answer against it. It already caught two real mistakes before any farmer saw them. When we're not sure, we ask one question or connect her to the Kisan Call Centre."

Tech. "React on the phone; Python and Flask on the server; Gnani Prisma and Timbre for speech; SQLite for the official registry; one Docker image on Vercel; 46 automated tests."

Close. "Next we pilot with farmers, add the problems they actually ask about, and get every entry verified by agriculture officers. Pantha Sahayam: speak your problem, hear the right answer."

11 Team & links

Member	Contribution
Name	e.g. product idea, farmer research, Telugu content, testing
Name	e.g. backend, Gnani integration, data pipeline
Name	e.g. frontend, design, demo video

Link	Address
Live app	https://....vercel.app
Source code	https://github.com/ghosttech07/Farmer-app
Demo video	link

LinkedIn / X post

🌾 We built Pantha Sahayam (పంట సహాయం) for the #GreatIndianAIInternshipChallenge by Gnani AI!

A small farmer shouldn't have to type in English to find out what to spray. With Pantha Sahayam they just SPEAK their crop problem in Telugu, Hindi or English, and hear back the right registered pesticide, the dose per litre and the safe waiting period before harvest.

🎙️ Gnani Prisma understands the farmer's voice
🔊 Gnani Timbre speaks the answer back, and the language can be switched anytime
📚 Every dose comes from the Government of India's CIB&RC registry (2,243 official uses), checked automatically
🧭 Vague question? It asks which crop, then guides to the answer
🚫 No login, no install, just a link

AI listens and speaks; it never invents a pesticide dose.

Try it: [live link]
Code: github.com/ghosttech07/Farmer-app

#GnaniAI #GreatIndianAIInternshipChallenge #AgriTech #VoiceAI #IndianLanguages